Roof Lifecycle Cost Tool: User Guide

The tool shows a building owner what their roof will cost over the next 20 years in two scenarios: with **PaxSeal planned maintenance**, or with **reactive repair** (fixing problems only after they leak). It turns the difference into dollars, using real cost and storm data for the owner's market.

Two kinds of cost curve

1

Generic cost curves

No inputs. The average cost of a roof from new to replacement, with and without PaxSeal, for every market, building type and roof type.

noah-austin.github.io/TOC-Model/generic

2

Custom cost curves

Enter a customer's roof in the builder and get a branded two-page report to print, save as PDF, or send as a link.

noah-austin.github.io/TOC-Model

3

Library

35 ready-made custom reports: a typical building of each property type in each market. Open or download, no typing.

noah-austin.github.io/TOC-Model/library

Markets: Austin, San Antonio, Hampton Roads, Culbertson (N. Virginia), Laurel. **Property types:** Warehouse / Industrial, Office, Retail, Multi-Family, Mixed Use, Medical / Institutional, K-12 / Education.

Make a customer report in two minutes

1. **Customer.** Customer name, property, city and contact. Add your name, phone and email; they print in the report's "Next step" box.
2. **Roof.** Pick the market and property type, then roof area, age, roof system and condition. Add warranty years left and active leaks if you know them.
3. **Costs (optional).** Leave blank to use market defaults (shown in gray). Enter the real replacement cost, the PaxSeal price from your proposal, or a cap rate if you have them; the report recalculates instantly.
4. **Share. Print / PDF** saves the two-page report. **Copy report link** gives a link that opens the finished report without the input form.

Customer copy vs internal copy. Reports are customer-ready by default. Under **Model assumptions**, "Show source confidence" makes an **internal review copy** that rates each data source and is marked "Internal review copy" on every page. Don't send that version to customers. The Model assumptions section is for sales ops only.

What's in a report

Page 1. The projected 20-year savings in one sentence, with its range; the property at a glance; net cost for each scenario; the cumulative spending chart; and a roof-life timeline showing when each scenario needs a new roof.

Page 2. Where the savings come from (fees, repairs, storm damage, replacements); up to four takeaways for this roof; the local storm risk with FEMA ratings; the key assumptions and their sources; and a next step with your contact details.

Finding your way

Every page has the same header: **Generic cost curves**, **Custom cost curves** (the report builder), **Library**, **Assumptions** and the **User guide**. The two kinds of curve are kept separate: generic curves are research averages with no inputs; custom curves are for one customer's roof. Every report has a link with its inputs built in. A report link sent to a customer shows only the report, with Edit and Print. The **Assumptions** page lists every number the model uses with its source; sales ops can adjust values there and publish them by uploading the downloaded overrides.json to the project.

How it works

1. Describe the roof

Market, property type, size, age, condition and roof system. Anything left blank uses a market default.

2. Look up market data

Local replacement cost per sq ft, county storm risk from the FEMA National Risk Index, roofing cost growth (4.6% a year, federal price index), and local cap rates by property type.

3. Simulate 20 years, both ways

Each year's fee, repairs, storm damage and any roof replacement is added up for both scenarios. That running total is the chart on page 1.

4. Compare

The difference in net 20-year cost is the projected savings, shown with a conservative-to-upside range.

	WITH PAXSEAL	REACTIVE REPAIR
Roof life	Full design life (20 yrs for TPO, EPDM, mod bit, BUR; 40 for metal)	About 5 years shorter
Program fee	Paid every year, rising 3% a year	None
Repairs	Caught early on scheduled visits; the fee covers half of planned repairs	About 3× the planned rate (more in hospitals, schools and apartments, where leaks do more damage)
Storm damage	Expected loss for the county	1.5× higher with reactive repair only
Replacement	Later	Sooner, and at a higher price because roofing costs keep rising

> **Net 20-year cost = cash spent + value of the current roof used up − roof life left at year 20**

Counting roof life left at the end keeps the comparison fair: a scenario that buys a new roof in year 19 gets credit for the 19 years it still has. The current roof's value is the same in both scenarios, so it never changes the savings.

Worked example

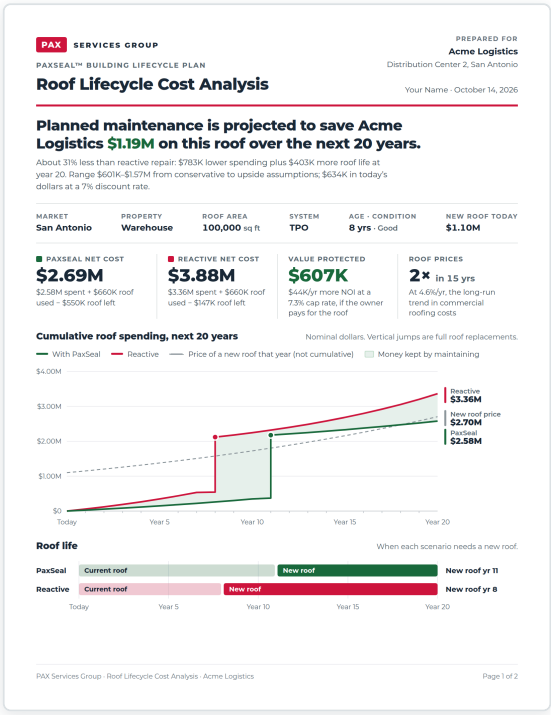
San Antonio warehouse: 100,000 sq ft TPO roof, 8 years old, good condition. A new roof costs about **$1.10M** today.

	PAXSEAL	REACTIVE
Next new roof	Year 11	Year 8
PaxSeal fees	$371K	—
Repairs	$268K	$1.58M
Storm damage	$138K	$207K
Roof replacements	$1.80M	$1.58M
Net 20-year cost	**$2.69M**	**$3.88M**

PROJECTED SAVINGS

$1.19M (31%)

$783K less cash spent, plus $403K more roof life at year 20. Range $601K–$1.57M. Property value protected: $607K (about $44K/yr better NOI ÷ 7.3% cap rate).


PAX SERVICES GROUP

PAXSEAL™ BUILDING LIFECYCLE PLAN

PREPARED FOR
Acme Logistics
Distribution Center 2, San Antonio

Roof Lifecycle Cost Analysis

Your Name · October 14, 2026

Planned maintenance is projected to save Acme Logistics $1.19M on this roof over the next 20 years.

About 31% less than reactive repair: $783K lower spending plus $403K more roof life at year 20. Range $601K–$1.57M from conservative to upside assumptions; $634K in today's dollars at a 7% discount rate.

MARKET	PROPERTY	ROOF AREA	SYSTEM	AGE · CONDITION	NEW ROOF TODAY
San Antonio	Warehouse	100,000 sq ft	TPO	8 yrs · Good	$1.10M

PAXSEAL NET COST	REACTIVE NET COST	VALUE PROTECTED	ROOF PRICES
$2.69M	$3.88M	$607K	2× in 15 yrs
$2.58M spent + $660K roof used − $550K roof left	$3.36M spent + $660K roof used − $147K roof left	$44K/yr more NOI at a 7.3% cap rate, if the owner pays for the roof	At 4.6%/yr, the long-run trend in commercial roofing costs

Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.

With PaxSeal · Reactive · Price of a new roof that year (not cumulative) · Money kept by maintaining

$4.00M
$3.00M
$2.00M
$1.00M
$0

Today · Year 5 · Year 10 · Year 15 · Year 20

Reactive $3.36M
New roof price $2.70M
PaxSeal $2.58M

Roof life

When each scenario needs a new roof.

PaxSeal: Current roof · New roof · New roof yr 11
Reactive: Current roof · New roof · New roof yr 8

Today · Year 5 · Year 10 · Year 15 · Year 20

PAX Services Group · Roof Lifecycle Cost Analysis · Acme Logistics

Page 1 of 2

Page 1 of this example report

Reading the report

ON THE REPORT	WHAT IT MEANS
Projected savings	How much less the roof costs over 20 years with PaxSeal, split into lower cash spending and extra roof life left at year 20.
Range	The same projection under cautious and optimistic assumptions for repair costs and roof life added.
Today's dollars	The savings discounted at 7% a year: what that future money is worth now.
Value protected	For investor-owned property: yearly operating savings ÷ cap rate. Applies when the owner pays for the roof (not a triple-net lease).
Repair & storm savings	Shown instead for schools and hospitals, which are usually owner-occupied: unbudgeted repair and storm costs avoided.
The chart	Cumulative spending for each scenario. Vertical jumps are roof replacements; the green band is the money kept by maintaining. The dashed line is the price of a new roof that year.
What this means	Up to four takeaways chosen for this roof: fewer roofs or a later replacement, smaller repairs, warranty, and property-type points.

Where the numbers come from

INPUT	SOURCE	STRENGTH
Storm risk	FEMA National Risk Index, December 2025, by county	STRONG
Roofing cost growth	BLS Producer Price Index for nonresidential roofing, 2007–2025	STRONG
Roof life with maintenance	Industry service-life tables; NRC Canada roof study; manufacturer warranty terms	MODERATE
Local roof prices, cap rates	Contractor and broker figures from search summaries; need verification	MODERATE
Repair cost multiplier, base repair rate	Industry estimates; no published study. These drive most of the savings, which is why the report shows a range	WEAK
PaxSeal fee	Estimate ($800 + $0.13/sq ft a year) until you enter the proposal price	Our price

When a customer pushes back

"Those savings look high."
Point to the range and to the split between cash and roof value. Even the conservative case is usually clearly positive. The biggest driver is buying fewer or later roofs.

"Is the PaxSeal fee included?"
Yes. It is a cost every year, rising 3% a year. Enter the actual proposal price for an exact figure.

"Doesn't insurance cover storms?"
Storm figures are before insurance, and the report says so. Storm damage is a small part of the savings; most comes from repairs and replacements.

"Our roof is different."
Enter the real replacement cost, fee or cap rate and the report recalculates on the spot.

Good to know

Estimates, not quotes. Every report says so. The PaxSeal proposal sets the actual price and scope.

Typical buildings in the library are illustrative profiles (for example, a 100,000 sq ft warehouse roof, 8 years old), not market averages. Use a custom report for a real property.

Updates. The tool is a single web page hosted on GitHub Pages and updates automatically when the model or data change. The full method is in **docs/MODEL_SPEC.md** in the project repository.